Hospital Readmission Risk Analysis

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I. Abstract

This study investigates the relationship between hemoglobin A1c (HbA1c) levels and early hospital readmission among patients with diabetes-related and other diagnoses. Using a dataset of nearly 70,000 inpatient encounters, I performed exploratory analysis and multivariable logistic regression to examine how HbA1c interacts with demographic, clinical, and administrative factors. While HbA1c was not a significant independent predictor of early readmission, interaction effects revealed its importance within specific subgroups, particularly patients with diabetes or injury-related admissions. Key predictors of early readmission included discharge disposition, admitting provider specialty, age, length of stay, and primary diagnosis. The final model, which incorporated key interactions, achieved an AUC of 0.618, indicating modest discrimination. These findings highlight the nuanced role of HbA1c and suggest that interventions to reduce early readmission should consider both clinical context and patient characteristics.

II. Introduction

Hospital readmission within 30 days is a critical quality and cost metric in healthcare, particularly for patients with chronic conditions like diabetes. Hemoglobin A1c (HbA1c) is widely recognized as a biomarker of long-term glucose control and is frequently used to assess the quality of diabetes care. The present analysis leverages a large clinical database of hospitalized patients with diabetes in the United States to evaluate whether HbA1c measurement — and clinical response to elevated HbA1c — is associated with early hospital readmission. Specifically, this study tests the hypothesis that the measurement of HbA1c, and associated medication adjustments in response to poor control (HbA1c > 8.0%), are associated with reduced readmission risk.

This study seeks to evaluate the relationship between HbA1c and early hospital readmission using a large administrative dataset. In addition to examining the standalone association between HbA1c and readmission, I assess how this relationship varies by diagnosis category and admitting specialty. Control variables include age, gender, race, admission source, discharge disposition, diagnosis, medical specialty, and length of stay. Our goal is to identify whether HbA1c contributes meaningfully to prediction and to understand its role in the broader context of readmission risk.

III. Data Set and Preparation

The data were obtained from the diabetic.data.final.csv file, which includes 69,973 patient encounters and 58 variables. Variables relevant to this analysis included:

- Outcome: Early readmission (binary indicator for readmission within 30 days)
- Primary predictor: HbA1c level, categorized as "High-change," "High-no change," "Normal," or "None"
- Control variables: Age (grouped), gender, race, admission source, discharge disposition, primary diagnosis, admitting specialty, and time spent in hospital

To prepare the dataset for modeling:

- Factor variables were re-leveled to ensure appropriate reference groups

- HbA1c was recoded into a categorical variable with meaningful clinical groupings
- Primary diagnoses were grouped into 9 broader categories to reduce sparsity
- Age was grouped into three bands: under 30, 30–60, and over 60
- Interaction terms were created based on exploratory findings and added iteratively

Observations with missing or implausible values in key predictors were removed. Logistic regression models were then fitted using the glm() function in R, with model selection guided by both theory and likelihood ratio tests. AUC and classification thresholds were evaluated using the pROC package.

IV. Methods

I conducted a retrospective observational analysis using a clinical dataset of 69,973 inpatient encounters from a U.S. hospital system. The dataset was originally assembled to evaluate historical patterns of diabetes care and includes detailed information on demographics, diagnoses, procedures, provider specialties, medications, lab results, and readmission history. The primary outcome was early readmission, defined as a binary indicator equal to 1 if a patient was readmitted within 30 days of discharge and 0 otherwise (including no readmission or readmission after 30 days). This 30-day threshold aligns with hospital quality benchmarks and funding agency criteria.

The primary predictor of interest was hemoglobin A1c (HbA1c), a clinical marker of long-term glucose control. HbA1c was categorized into four clinically meaningful groups: (1) no test performed, (2) test performed with a normal result, (3) result >8% with no change in diabetic medications, and (4) result >8% with a change in medications. Medication change was defined as any inpatient adjustment in dosage or switch to a different drug class. In rare cases (<0.1%), a prior outpatient HbA1c result from within three months of admission was used in place of a repeat inpatient test.

Additional covariates included age (grouped into <30, 30–60, and >60 years), gender, race, admission source, discharge disposition (home vs. otherwise), primary diagnosis (grouped into nine categories), admitting provider specialty, and time in hospital. Exploratory analyses involved cross-tabulations, chi-square tests, and t-tests to examine univariate associations with early readmission.

Multivariable modeling was conducted using logistic regression via the `glm()` function in R. A base model included only control variables, followed by a model including HbA1c. Pairwise interactions between HbA1c and key covariates (e.g., diagnosis, specialty) were tested using likelihood ratio tests to assess potential effect modification. Additional interactions among provider, diagnosis, and discharge variables were also examined. Final model selection was based on statistical significance and theoretical relevance. Model performance was assessed using deviance, Akaike Information Criterion (AIC), and area under the Receiver Operating Characteristic curve (AUC). The optimal classification threshold was identified using Youden's J statistic.

V. Results

Exploratory Analysis

Exploratory analyses revealed several patient characteristics and clinical factors associated with early hospital readmission. HbA1c levels were unevenly distributed, with the majority of patients having no recorded value (81.6%). A chi-square test showed that HbA1c status was significantly associated with early readmission ($p = 0.021$), although the differences in readmission rates across HbA1c categories were modest (ranging from 7.4% to 9.1%).

Age was a strong predictor: patients aged 60 and older had a significantly higher readmission rate (9.9%) compared to those under 30 (6.2%) ($p < 2.2e-16$). Other significant predictors included race ($p = 0.019$), admission source ($p = 0.022$), and discharge disposition ($p < 2.2e-16$). Notably, patients discharged to a location other than home had a much higher readmission rate (12.5%) than those discharged home (6.9%).

Primary diagnosis and admitting specialty were also associated with early readmission. Diagnosis categories such as injury and diabetes had elevated readmission rates, while respiratory and digestive conditions had lower rates ($p = 3.6e-13$). Among providers, family/general practice and internal medicine were associated with the highest readmission rates ($p = 1.6e-08$). Additionally, patients readmitted early stayed longer in the hospital (mean = 4.8 days vs. 4.2 days; $p < 2.2e-16$).

Base Predictive Model (Control Variables Only)

A logistic regression model including control variables identified significant predictors of early readmission. Key findings:

- Discharge disposition had the strongest effect: patients not discharged home were significantly more likely to be readmitted ($p < 0.001$).
- Older age (60+) was significantly associated with higher readmission risk ($p = 0.0029$).
- Admission source (Other) was associated with lower readmission odds ($p < 0.001$).
- Time in hospital was positively associated with readmission ($p < 0.001$).
- Some diagnosis categories (e.g., respiratory, musculoskeletal) were associated with reduced odds of readmission, while certain admitting specialties (e.g., internal medicine, F-GP) were associated with increased odds.

HbA1c was not included in this base model, providing a foundation for evaluating its added contribution in subsequent steps.

Adding HbA1c to the Model

Introducing HbA1c into the model produced a marginally significant improvement in model fit (likelihood ratio test: $\chi^2 = 7.38$, $df = 3$, $p = 0.061$). However, none of the HbA1c categories emerged as significant main effect predictors in the adjusted model. This suggests that HbA1c alone does not independently explain early readmission risk beyond the effect of other clinical and administrative factors.

Testing Pairwise Interactions

A series of likelihood ratio tests assessed potential two-way interactions:

- Significant interactions included:
  - admit.spec × discharge_disposition ($p = 0.012$)
  - prim.diag × admit.spec ($p = 0.001$)
- Borderline significant: age3 × prim.diag ($p = 0.054$)
- Not significant: admission.source × discharge_disposition, age3 × time_in_hospital

These results suggest that how diagnoses and discharge locations affect readmission risk may depend on the type of provider managing the patient's care.

HbA1c Interactions in the Final Model

Further model refinement tested interactions between HbA1c and other variables. Two interactions significantly improved model fit:

- HbA1c × Primary Diagnosis ($p = 0.002$)
- HbA1c × Admitting Specialty ($p = 0.034$)

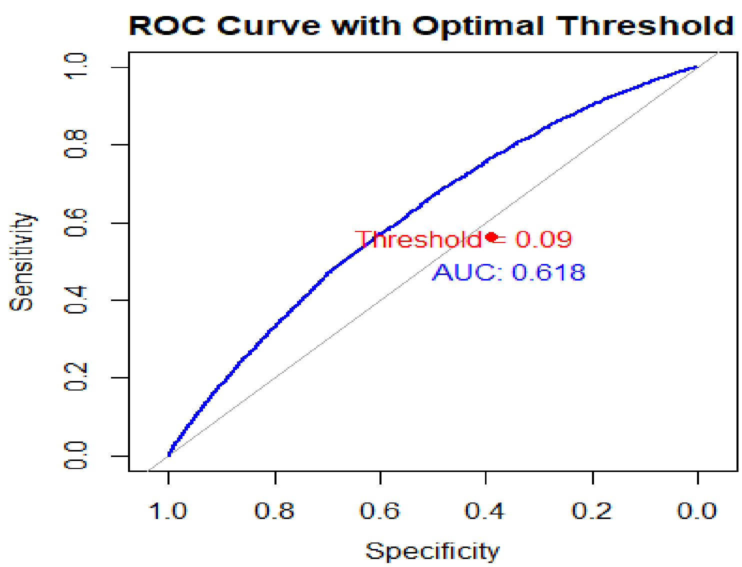
This suggests the effect of HbA1c on early readmission is context-dependent, varying by diagnosis and provider type. For example, patients admitted for diabetes or injury with normal or missing HbA1c values had higher odds of early readmission compared to those with elevated levels. This highlights that HbA1c's utility is not as a main effect, but rather as a modifying factor within subgroups.

Model Fit and Performance

The final model included main effects and interactions among HbA1c, diagnosis, specialty, discharge status, age, admission source, and hospital stay length. Key fit metrics:

- Residual deviance: 39,552 (from null deviance 40,490)
- AIC: 39,766
- AUC: 0.618 — indicates modest discrimination ability
- Optimal threshold: 0.0897 (Youden's J)
    - Sensitivity: 56.2%
    - Specificity: 60.9%

Using the optimized threshold improved classification performance compared to the default 0.5 cutoff, which failed to predict any early readmissions.



VI. Discussion

This study evaluated the role of HbA1c and other clinical and administrative variables in predicting early hospital readmission among patients with diabetes and related conditions. Exploratory analyses revealed that several factors were significantly associated with readmission, including age, primary diagnosis, provider specialty, discharge disposition, admission source, and time spent in the hospital. These variables informed the development of a multivariable logistic regression model that adjusted for both demographic and care-related covariates.

While HbA1c alone did not emerge as a significant independent predictor of early readmission, its inclusion modestly improved model fit. More importantly, interaction terms revealed that the association between HbA1c and readmission varied significantly across diagnosis categories and admitting specialties. For example, among patients admitted for diabetes or injury, normal or missing HbA1c values were associated with higher readmission odds — particularly when there was no corresponding adjustment to diabetic medications. These findings suggest that HbA1c is most informative when interpreted in clinical context, rather than as a standalone predictor.

This analysis also confirmed the strong influence of discharge disposition and provider specialty on early readmission risk. Patients not discharged to home had nearly double the odds of readmission, and interactions between provider specialty and diagnosis indicated that the same condition may confer different risks depending on who manages the patient's care. These findings highlight the importance of care coordination, discharge planning, and provider-specific practices in reducing preventable readmissions.

Importantly, this study was motivated by the hypothesis that HbA1c measurement — and particularly provider response to elevated HbA1c — would reduce early readmission. Although no main effect was observed, our findings support the idea that provider behavior in response to HbA1c values (i.e., medication adjustments) may be a critical link between measurement and outcomes. This reinforces the value of HbA1c not just as a passive indicator, but as a potential trigger for action in managing high-risk patients.

While the model achieved modest discrimination (AUC = 0.618), using an optimized threshold improved sensitivity to 56% while maintaining specificity at 61%. This suggests the model could serve as a screening tool to identify patients at elevated risk for early readmission, particularly when HbA1c context and provider actions are considered.

VII. Conclusion

This analysis found that while HbA1c levels do not independently predict early hospital readmission, they do have significant effects within certain clinical contexts — particularly among patients with diabetes and injury-related diagnoses. Other variables such as discharge disposition, admitting specialty, primary diagnosis, and hospital stay duration were stronger predictors of readmission across the full sample. Interaction terms involving HbA1c revealed that its importance may be masked when examined in isolation. These findings suggest that early readmission is influenced by both patient condition and provider-level factors, and that predictive modeling should account for such interactions. Improving HbA1c documentation, especially for high-risk diagnoses, and tailoring discharge planning by specialty could enhance efforts to reduce avoidable readmissions.

VIII. Appendix (Code)

```
#### Question 1 ####
# Load libraries
library(tidyverse)

# Load data
df <- read.csv("diabetic.data.final.csv", stringsAsFactors = FALSE)

# Quick peek
str(df)
summary(df)

### As Factor ###
df$gender <- as.factor(df$gender)
df$race <- as.factor(df$race)
df$age3 <- as.factor(df$age3)
df$admission.source <- as.factor(df$admission.source)
df$discharge_disposition <- as.factor(df$discharge_disposition)
df$prim.diag <- as.factor(df$prim.diag)
df$admit.spec <- as.factor(df$admit.spec)
df$HbA1c <- as.factor(df$HbA1c)
df$early.readmit <- as.factor(df$early.readmit)


#### HbA1c ####
table(df$HbA1c)
prop.table(table(df$HbA1c))

table(df$HbA1c, df$early.readmit)
prop.table(table(df$HbA1c, df$early.readmit), margin = 1)

chisq.test(table(df$HbA1c, df$early.readmit))


#### Control Variables ####
#Gender
table(df$gender, df$early.readmit)
chisq.test(table(df$gender, df$early.readmit))

# Race
table(df$race, df$early.readmit)
chisq.test(table(df$race, df$early.readmit))

# Age
table(df$age3, df$early.readmit)
chisq.test(table(df$age3, df$early.readmit))
```

```
# Admission Source
table(df$admission.source, df$early.readmit)
chisq.test(table(df$admission.source, df$early.readmit))

# Discharge Dispo
table(df$discharge_disposition, df$early.readmit)
chisq.test(table(df$discharge_disposition, df$early.readmit))

# Primary Diagnosis
table(df$prim.diag, df$early.readmit)
chisq.test(table(df$prim.diag, df$early.readmit))

# Medical Specialty
table(df$admit.spec, df$early.readmit)
chisq.test(table(df$admit.spec, df$early.readmit))

# Time in Hospital
tapply(df$time_in_hospital, df$early.readmit, summary)
t.test(time_in_hospital ~ early.readmit, data = df)

### Question 2 ###
# 1. Base model WITHOUT HbA1c
base_model <- glm(
  early.readmit ~ age3 + admission.source + discharge_disposition +
    prim.diag + admit.spec + time_in_hospital,
  data = df_model,
  family = binomial
)

# 2. View the summary of the base model
summary(base_model)

### Question 3 ###
# 3. Now fit the full model WITH HbA1c added back
hbA1c_model <- glm(
  early.readmit ~ HbA1c + age3 + admission.source + discharge_disposition +
    prim.diag + admit.spec + time_in_hospital,
  data = df_model,
  family = binomial
)

# 4. View the summary with HbA1c
summary(hbA1c_model)

# 5. Perform a likelihood ratio test to see if adding HbA1c improves model fit
```

```r
anova(base_model, hbA1c_model, test = "LRT")

### Question 4 ###
# 1. age3 × prim.diag
interact1 <- glm(early.readmit ~ age3 * prim.diag + admission.source +
                  discharge_disposition + admit.spec + time_in_hospital,
                data = df_model, family = binomial)
anova(base_model, interact1, test = "LRT")

# 2. admission.source × discharge_disposition
interact2 <- glm(early.readmit ~ age3 + admission.source * discharge_disposition +
                  prim.diag + admit.spec + time_in_hospital,
                data = df_model, family = binomial)
anova(base_model, interact2, test = "LRT")

# 3. admit.spec × discharge_disposition
interact3 <- glm(early.readmit ~ age3 + admission.source + discharge_disposition *
                  admit.spec + prim.diag + time_in_hospital,
                data = df_model, family = binomial)
anova(base_model, interact3, test = "LRT")

# 4. age3 × time_in_hospital
interact4 <- glm(early.readmit ~ age3 * time_in_hospital + admission.source +
                  discharge_disposition + prim.diag + admit.spec,
                data = df_model, family = binomial)
anova(base_model, interact4, test = "LRT")

# 5. prim.diag × admit.spec
interact5 <- glm(early.readmit ~ age3 + admission.source + discharge_disposition +
                  prim.diag * admit.spec + time_in_hospital,
                data = df_model, family = binomial)
anova(base_model, interact5, test = "LRT")


# Example: age3 × prim.diag
interact_model1 <- glm(
  early.readmit ~ age3 * prim.diag + admission.source + discharge_disposition +
    admit.spec + time_in_hospital,
  data = df_model,
  family = binomial
)

# Compare to base model
anova(base_model, interact_model1, test = "LRT")
```

```
### Question 5 ###
# 1. HbA1c × age3
hbA1c_inter1 <- glm(early.readmit ~ HbA1c * age3 + admission.source +
                    discharge_disposition + prim.diag + admit.spec +
                    time_in_hospital,
                  data = df_model, family = binomial)
anova(hbA1c_model, hbA1c_inter1, test = "LRT")

# 2. HbA1c × prim.diag
hbA1c_inter2 <- glm(early.readmit ~ HbA1c * prim.diag + age3 + admission.source +
                    discharge_disposition + admit.spec + time_in_hospital,
                  data = df_model, family = binomial)
anova(hbA1c_model, hbA1c_inter2, test = "LRT")

# 3. HbA1c × discharge_disposition
hbA1c_inter3 <- glm(early.readmit ~ HbA1c * discharge_disposition + age3 +
                    admission.source + prim.diag + admit.spec + time_in_hospital,
                  data = df_model, family = binomial)
anova(hbA1c_model, hbA1c_inter3, test = "LRT")

# 4. HbA1c × admit.spec
hbA1c_inter4 <- glm(early.readmit ~ HbA1c * admit.spec + age3 + admission.source +
                    discharge_disposition + prim.diag + time_in_hospital,
                  data = df_model, family = binomial)
anova(hbA1c_model, hbA1c_inter4, test = "LRT")

# 5. HbA1c × time_in_hospital
hbA1c_inter5 <- glm(early.readmit ~ HbA1c * time_in_hospital + age3 + admission.source +
                    discharge_disposition + prim.diag + admit.spec,
                  data = df_model, family = binomial)
anova(hbA1c_model, hbA1c_inter5, test = "LRT")

### Question 6 ###
## Final Model ##
final_model <- glm(
  early.readmit ~ HbA1c * prim.diag +
    HbA1c * admit.spec +
    admit.spec * discharge_disposition +
    prim.diag * admit.spec +
    age3 + admission.source + time_in_hospital,
  data = df_model,
  family = binomial
)
summary(final_model)

library(pROC)
```

```
# 1. Create the ROC object
roc_obj <- roc(df_model$early.readmit, pred_probs)

# 2. Find optimal threshold using Youden's J
optimal_coords <- coords(roc_obj, x = "best", best.method = "youden", ret = c("threshold",
"sensitivity", "specificity"), transpose = FALSE)
optimal_threshold <- optimal_coords$threshold
optimal_sens <- optimal_coords$sensitivity
optimal_spec <- optimal_coords$specificity

# 3. Plot the ROC curve
plot(roc_obj, col = "blue", main = "ROC Curve with Optimal Threshold", print.auc = TRUE)

# 4. Add point for optimal threshold
points(1 - optimal_spec, optimal_sens, col = "red", pch = 19)
text(1 - optimal_spec + 0.05, optimal_sens, labels = paste0("Threshold = ",
round(optimal_threshold, 3)), col = "red")
```